Enterprise Full-Stack Project Instructions

Final Project Objective

• A fully operational PostgreSQL or MSSQL database

• A structured back-end API connected directly to your database

• Complete API architecture with controllers, services, repositories/data access, models/DTOs

• API must return live database results

• Full Docker containerization for database and API

• CI/CD pipelines using GitHub for PR validation and merge deployment

• Deployment to AWS cloud infrastructure

• Professional technical documentation

• Final team presentation

Project Parts

• Part 1 — Team Implementation: Team roles, GitHub repo, Trello board, Agile workflow

• Part 2 — Planning & Architecture: Functional/non-functional requirements, diagrams, deployment planning

• Part 3 — Database Development: Local DB setup, schema, relationships, seed data

• Part 4 — API Development: RESTful API, CRUD, architecture, testing

• Part 5 — Containerization + CI/CD + Cloud: Docker, GitHub Actions, AWS deployment

• Part 6 — Testing, Documentation & Presentation

Agile Sprint Breakdown

• Sprint 1: Project Planning

• Sprint 2: Documentation & Architecture

• Sprint 3: Database Development

• Sprint 4: API Development

• Sprint 5: Containerization + CI/CD

• Sprint 6: AWS Deployment + Final Presentation

Required Deliverables

• GitHub Repository Link

• Trello Board Link

- PDF Documentation
- User Guide
- Architecture Guide
- Database Schema
- Docker Files
- CI/CD Workflow Files
- AWS Deployment Proof
- Final Presentation

Evaluation Criteria

- Planning quality
- Technical implementation
- API functionality
- Database design
- Containerization
- CI/CD implementation
- AWS deployment
- Team collaboration
- Documentation quality
- Presentation quality

Important Notes

• Projects must follow Agile methodology

• Scrum Master rotates weekly

• Daily meetings are expected

• Professional documentation is mandatory

• Enterprise-grade architecture is required

• This project simulates real-world software delivery